Multi-Dialect Marathi Text Normalization via LLM-Driven Synthetic Augmentation and Neural Sequence-to-Sequence Benchmarking

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Abstract. Low-resource regional dialects pose severe challenges for Natural Language Processing (NLP) systems due to significant morpho-syntactic variation, non-standard orthography, and extreme scarcity of annotated parallel corpora. In this paper, we address the task of dialect normalization—converting non-standard regional dialectal text into Standard Written Marathi—focusing on three major Indo-Aryan regional dialects of Maharashtra: Malvani (D1), Ahirani (D2), and Varhadi (D4). To overcome severe data scarcity, we introduce an automated synthetic parallel data expansion pipeline combining large language model (LLM) generation using Gemma-2 (9B/27B) with a multi-tier verification engine (rule-based script filtering and LLM semantic auditing), effectively doubling the clean parallel training corpus from 16,163 to 32,335 verified pairs across agriculture and finance domains. We conduct a rigorous empirical evaluation of state-of-the-art multilingual sequence-to-sequence transformer architectures—google/mT5-small (300M), IndicBART (244M), and AI4Bharat IndicTrans2 (320M)—under an 85/15 stratified test split with 5-fold cross-validation across 8 dataset configurations. Empirical results demonstrate that mT5-small achieves state-of-the-art performance with 69.51 BLEU and 84.58 chrF++ on the 32k combined multi-dialect dataset (+12.39 BLEU over IndicBART), with individual dialect performance reaching 80.99 BLEU and 91.21 chrF++ on Varhadi (D4). We present detailed architectural comparisons, data scaling ablation studies, and qualitative subword morphological error analyses, establishing a benchmark for Indic dialect text normalization.

Keywords: Dialect Normalization · Low-Resource NLP · Marathi Dialects · Sequence-to-Sequence Modeling · Synthetic Data Augmentation · mT5 · IndicBART · Text Standardization.

1 Introduction

Natural Language Processing (NLP) has achieved remarkable advances across high-resource languages in recent years, driven by the emergence of large-scale pre-trained Transformer models [24,21]. However, a significant digital divide persists: the vast majority of the world's 7,000+ languages remain critically underserved in computational resources, annotated datasets, and pre-trained model coverage. Among these, regional dialects of major languages present a particularly nuanced challenge—they are spoken by millions of people in everyday communication but are rarely represented in formal written corpora, standardized digital lexicons, or training data for downstream NLP applications.

Marathi, an Indo-Aryan language spoken by over 83 million native speakers primarily in the state of Maharashtra, India, exemplifies this challenge. While Standard Marathi (based on the Pune/Deshi variety) possesses a well-established literary tradition and growing digital presence, regional varieties—including Malvani (spoken across the Konkan coastal belt), Ahirani (spoken in the Khandesh region), and Varhadi (spoken in Vidarbha/Eastern Maharashtra)—exhibit profound phonological, lexical, and morphological divergences from the standard form [11,13,23,22]. These dialects are the primary means of oral communication for tens of millions of rural speakers, particularly in agriculture and finance domains, yet they remain virtually invisible to mainstream NLP infrastructure.

The practical consequences of this gap are severe. Downstream applications such as Machine Translation (MT), Information Retrieval, sentiment analysis, speech transcription post-processing, and conversational AI fail when exposed to raw dialectal inputs due to high Out-Of-Vocabulary (OOV) rates, altered verbal inflections, non-standard orthographic conventions, and informal phonetic spellings.

Dialect Normalization—the automated transformation of non-standard dialectal text into standard written orthography while preserving underlying semantic intent—serves as an essential pre-processing pipeline for bridging this digital divide [15]. By mapping dialectal surface forms to their standardized equivalents, normalization enables existing NLP models trained on Standard Marathi to process dialectal inputs without requiring expensive retraining or domain-specific architectural modifications.

1.1 Linguistic Background and Devanagari Orthographic Heterogeneity

Devanagari, an abugida writing system used for Marathi, Hindi, and Sanskrit, exhibits inherent orthographic complexities. While Standard Marathi possesses formal orthographic rules codified by the *Maharashtra Sahitya Parishad* (governing rules for *anusvara* nasalization, *hrasva-dirgha* vowel length, and consonant cluster representation), regional dialects operate almost exclusively in oral modalities. When speakers express dialectal forms in written digital media (e.g., social messaging, user queries, transcription post-processing), they use informal phonetic spellings. This introduces several distinct layers of orthographic heterogeneity:

– **Vocalic Length Diphthong Alternations**: Informal dialectal text frequently conflates short (*hrasva*) and long (*dirgha*) vowels (e.g., इ/ई and उ/ऊ), or substitutes monophthongs for diphthongs (e.g., Standard होता → Malvani हतो).

- **Nasalization Anusvara Variations**: Standard Marathi strictly regulates *anusvara* placement for nasal consonants and grammatical inflections. Dialectal text frequently omits *anusvaras* or converts them to explicit nasal consonants (न, म, ङ).
- **Nukta Retroflex Consonant Shifts**: Dialects like Ahirani systematically simplify retroflex consonants (ळ → ल or य) and alter dental-palatal fricative conventions (श/ष → स), leading to extreme out-of-vocabulary (OOV) rates in standard tokenizers.
- **Agglutinative Postposition Merging**: Standard Marathi attaches case markers (ला, ने, ची, त) directly to modified stem nouns (सामान्यरूप). Dialects modify the stem in non-standard ways (e.g., Standard मुलाला → Ahirani पोराले, Malvani चेडवाक).

However, developing robust deep learning models for dialect normalization encounters two fundamental bottlenecks:

1. **Extreme Data Scarcity**: Publicly available parallel corpora mapping regional Indic dialects to their standard written forms are virtually non-existent. Unlike well-resourced dialect pairs (e.g., Swiss German to Standard German), Marathi dialectal variation has received minimal attention in computational linguistics, resulting in a near-total absence of curated dialect↔standard parallel datasets.
2. **Morphological Complexity**: Marathi is a morphologically rich, agglutinative language with complex inflectional paradigms. Dialectal variation modifies not only lexical items but also case markers, postpositions, and verbal tense suffixes in systematic but context-dependent ways. This requires models to learn fine-grained subword-level grammatical transformations rather than relying on naive word-level dictionary lookups.

To address these challenges, this study presents a systematic, end-to-end framework for multi-dialect Marathi text normalization. Our key contributions are:

- **LLM-Driven Synthetic Data Augmentation Pipeline**: We introduce an automated parallel dataset generation architecture leveraging Gemma-2 (9B/27B) [8] paired with multi-tier LLM-based verification and closed-loop correction, doubling clean parallel pairs from 16,163 to 32,335 across three Marathi dialects while maintaining strict semantic fidelity.
- **Rigorous Benchmark Evaluation**: We implement a standardized 85/15 stratified train-test split combined with 5-fold cross-validation across 8 distinct dataset configurations, evaluating three representative multilingual sequence-to-sequence transformer architectures: IndicBART (244M) [3], google/mT5-small (300M) [26], and AI4Bharat IndicTrans2 (320M) [6].
- **State-of-the-Art Results**: mT5-small achieves 69.51 BLEU [20] and 84.58 chrF++ on the 32k combined multi-dialect dataset (+12.39 BLEU over IndicBART), with individual dialect performance reaching 80.99 BLEU and 91.21 chrF++ for Varhadi (D4).

– **Architectural & Qualitative Insights**: We provide an in-depth analysis of vocabulary coverage, positional encoding mechanisms, and subword morphological error patterns across dialects, establishing a clear difficulty hierarchy (Varhadi $<$ Ahirani $<$ Malvani).

The remainder of this paper is organized as follows: Section 2 surveys related work in dialect normalization, Indic NLP, and synthetic data generation. Section 3 details our dataset construction pipeline, synthetic expansion architecture, and neural model setups. Section 4 presents quantitative benchmark results, ablation studies, and qualitative predictions. Section 5 concludes with a summary and directions for future research.

2 Related Work

2.1 Dialect Normalization and Text Standardization

The task of normalizing dialectal text into a standardized written form has been formally established as a distinct sentence-level character transduction task alongside historical and user-generated content normalization [15]. Early neural explorations by Partanen et al. [18] evaluated sequence transduction across 23 Finnish dialect varieties, demonstrating that sliding-window chunk-level neural architectures (LSTMs and character Transformers) effectively normalize non-standard dialectal forms into normative Standard Finnish, drastically reducing word error rates from an initial 52.89% down to 5.73%. Kuparinen et al. [15] extended this paradigm to a large-scale multilingual benchmark covering four distinct language families (Finnish, Norwegian, Swiss German, and Slovene), showing that fine-tuned sequence-to-sequence transformers and pre-trained byte-level architectures (byT5) consistently surpass character-level statistical machine translation (SMT) and recurrent baselines across diverse dialectal corpora.

In the Arabic NLP community, the Conventional Orthography for Dialectal Arabic (CODA) framework [9] established foundational orthographic guidelines for mapping diverse regional Arabic varieties (Egyptian, Levantine, Gulf, Maghrebi) to Modern Standard Arabic (MSA), providing the linguistic substrate for parallel corpus creation and computational Arabic dialect processing. The MADAR project subsequently extended these principles to fine-grained city-level dialect identification and translation across 25 Arab cities. While early computational approaches relied on character-level finite-state transducers (FSTs) and phrase-based SMT rules, contemporary work predominantly leverages pre-trained multilingual encoder-decoder transformers.

In Germanic languages, Swiss German dialect normalization to Standard High German has been investigated extensively due to the total absence of a single standardized orthography for Swiss German dialects. Early studies explored character-level statistical rules, while recent work by Kresic and Abbas [12] demonstrated that fine-tuning multilingual sequence-to-sequence transformers on the SwissDial corpus yields high-quality text normalization, with the compact `mT5-small` model achieving competitive performance against significantly larger base and large transformer variants.

For Indic languages, dialect normalization remains a critically underexplored area. While prior literature has addressed code-mixing (e.g., Hindi-English) and script transliteration, mapping regional dialectal text to a standardized written form within the same language has received minimal computational attention. A comprehensive global survey by Joshi et al. [10] categorizes dialectal NLP tasks across Natural Language Understanding (NLU) and Natural Language Generation (NLG) across diverse language families, emphasizing the acute performance degradation of state-of-the-art models on non-standard dialects and calling for equitable language technologies. Recently, Dimakis et al. [4] introduced a hybrid normalization framework combining rule-based morphological transformations with LLM few-shot prompting to standardize low-resource Greek dialect proverbs without requiring prior parallel training data.

In the specific context of Marathi language dialect processing, research has historically concentrated on speech, acoustic classification, or localized applications:

- **Acoustic Speech Classification Recognition**: Mulla and Kumar [17] explored spectro-temporal feature extraction (15 parameters including chromagram, spectral centroid, and MFCCs) with Chi-Square/ANOVA feature selection on 7,555 speech recordings from the LDC-IL Marathi raw speech corpus across four dialects (Goa-based, Vidarbha, Marathwada, Puneri), achieving 84.24% accuracy with Ridge Classifiers. Pawar et al. [19] investigated acoustic dialect identification across Ahirani, Varhadi, and Standard Marathi audio using MFCC features and Convolutional Neural Networks (CNNs) under clean and artificially generated Gaussian noise conditions (SNR 1.78–9.74 dB), achieving up to 93.65% accuracy in clean environments and 87.29% under noisy conditions. Amartyaveer et al. [1] and Kumar et al. [14] at IISc Bangalore introduced RESPIN-S1.0 (a 10,000+ hour read-speech corpus across 38 Indian dialects and 9 languages, including 4 Marathi subdialects: D1 Southern Konkan, D2 Northern Konkan, D3 Standard Marathi, and D4 Varhadi) and developed multimodal ASR and RoBERTa embeddings for dialect identification, achieving 79.81% accuracy.
- **Text-Level Dialect Processing Translation**: Gaikwad et al. [5] explored zero-shot prompting via the commercial Gemini API within a Retrieval-Augmented Generation (RAG) framework using Sentence Transformers and FAISS to translate Konkani (Devanagari) dialect queries into Standard Marathi for document Q&A. Vyavhare et al. [25] proposed a hybrid rule-based dictionary and neural translation web interface for regional Marathi dialects. Gavali [7] introduced a student-level framework combining IndicBERT and IndicWav2Vec for dialect detection and normalization on small local datasets.

Our work significantly advances this landscape: while prior studies focused either on speech/acoustic classification or API-dependent zero-shot prompting without releasing parallel datasets, we establish the first large-scale 32,335-pair parallel text benchmark, introduce an automated Gemma-2 LLM synthetic data expansion engine with multi-tier verification, and conduct rigorous 5-fold cross-validated fine-tuning of open-weights sequence-to-sequence transformers (`mT5-small`, IndicBART, IndicTrans2) for local, reproducible inference.

2.2 Indic NLP and Sequence-to-Sequence Transformers

The development of pre-trained multilingual models for Indic languages has accelerated significantly in recent years. IndicBART [3] introduced a multilingual sequence-to-sequence model based on the mBART-50 architecture [16], pre-trained on 11 Indic languages using a unified Devanagari script representation with a 64,000-token SentencePiece vocabulary. The model leverages orthographic similarity between Indic scripts to facilitate cross-lingual transfer learning and demonstrated competitive performance on neural machine translation and extreme summarization tasks despite being significantly smaller than general-purpose multilingual models.

IndicTrans2 [6] scaled transformer-based translation architectures to cover all 22 scheduled Indian languages, introducing the Bharat Parallel Corpus Collection (BPCC) containing 230 million bitext pairs and the IN22 n-way parallel evaluation benchmark. The model employs script unification and optimized tokenization strategies to improve performance, particularly on low-resource language pairs.

Concurrently, general multilingual models such as mT5 [26]—the multilingual extension of T5 [21]—extended Text-to-Text Transfer Transformers across 101 languages using the mC4 (Multilingual Colossal Clean Crawled Corpus). mT5 features a 250,000-token SentencePiece vocabulary and demonstrated state-of-the-art performance on the XTREME multilingual benchmark. Its text-to-text formulation, which casts all NLP tasks as input-output text transformations, makes it particularly well-suited for sequence-level normalization tasks.

The L3Cube-MahaNLP initiative [11] has contributed Marathi-specific transformer models (MahaBERT, MahaRoBERTa) and curated datasets for tasks including sentiment analysis, named entity recognition, and hate speech detection. However, these encoder-only models are designed for classification tasks rather than sequence-to-sequence generation required for dialect normalization.

2.3 Synthetic Data Generation for Low-Resource NLP

The use of Large Language Models (LLMs) for synthetic data generation has emerged as a powerful strategy for overcoming data scarcity in low-resource settings. Recent work has demonstrated that LLMs function more effectively as data generators than as direct task solvers, enabling a "generate-then-fine-tune" paradigm where synthetic parallel data from a large teacher model is used to train smaller, task-specific student models [8]. Key considerations in this approach include maintaining semantic fidelity, preventing hallucination, and implementing multi-tier verification pipelines to ensure data quality [2].

For Indic languages specifically, synthetic data augmentation has been applied primarily to machine translation tasks, with back-translation and pivoting through high-resource languages serving as common strategies. Our work extends this paradigm specifically to the dialect normalization task, introducing domain-aware generation templates and multi-stage LLM verification protocols tailored to the morphological properties of Marathi dialects.

3 Methodology

3.1 Target Dialects and Linguistic Properties

We target three major Marathi dialects mapped in social dialectology and comparative linguistic surveys of Maharashtra [13,22] that collectively span the primary geographic regions of Maharashtra and exhibit distinct linguistic transformation patterns:

1. **Malvani (D1, Konkan Coast)**: Characterized by phonetic shortening of word-final vowels, palatalization of dental consonants, and distinct auxiliary verb forms. Lexical transformations include systematic vowel shifts: standard जातो ("he goes") becomes Malvani जातां; standard कुठे ("where") becomes कुठंय. Domain-specific terminology in agriculture and banking is generally preserved but often accompanied by Konkani-influenced function words.
2. **Ahirani (D2, Khandesh Region)**: Heavily influenced by neighboring Gujarati and Bhili language families, exhibiting significant consonant cluster simplification and retroflex-dental consonant shifts. Characteristic transformations include: standard कशाला ("why") → Ahirani कसाले; standard मुलगा ("boy") → Ahirani पोर. Ahirani presents the highest degree of lexical divergence among the three target dialects.
3. **Varhadi (D4, Vidarbha/Eastern Maharashtra)**: Spoken in Vidarbha, exhibiting characteristic interrogative transformations and systematic verb suffix modifications. Standard interrogative का? ("why?") becomes Varhadi काऊन?; verbal forms undergo regular suffix mapping patterns. Among the three dialects, Varhadi maintains the highest degree of structural similarity to Standard Marathi, with transformations being more systematic and rule-governed.

Table 1 provides a detailed comparative linguistic breakdown of representative sentences across the three target dialects alongside Standard Marathi, highlighting phonological, morphological, and lexical transformation paradigms.

3.2 Original Corpus Curation and Foundation in RESPIN

The foundational dialectal sentence prompts for our dataset were drawn from the domain-specific text transcripts of the RESPIN-S1.0 initiative [14], developed at IISc Bangalore under the Gates Foundation project. While RESPIN provides audio recordings and mono-lingual text prompts across agriculture and banking domains for Indian regional dialects (including Marathi sub-dialects: D1 Southern Konkan/Sindhudurg, D2 Northern Konkan/Ahirani, and D4 Varhadi/Nagpur),

Table 1. Comparative Linguistic Profile Sample Transformations Across Marathi Sub-Dialects

Dialect Region	Dialectal Input	Standard Marathi Output	Transformation Type
D1: Malvani	मका आज बँकेंत जावचा हा.	मला आज बँकेत जायचे आहे.	Dative pronoun (मका → मला), Infinitival verb (जावचा → जायचे), Auxiliary (हा → आहे).
D1: Malvani	तुमी खय चाललात?	तुम्ही कुठे चालला आहात?	Interrogative (खय → कुठे), Plural verb suffix (चाललात → चालला आहात).
D2: Ahirani	मना पोराले शेतीमा काम करणा शे.	माझ्या मुलाला शेतीत काम करायचे आहे.	Genitive pronoun (मना → माझ्या), Noun (पोराले → मुलाला), Locative (मा → त), Auxiliary (शे → आहे).
D2: Ahirani	तुम्ही कशाले इतला राग करताय?	तुम्ही कशाला एवढा राग करता?	Interrogative (कशाले → कशाला), Demonstrative (इतला → एवढा).
D4: Varhadi	तू काऊन बँकेंत गेला नव्हता?	तू का बँकेत गेला नव्हतास?	Interrogative (काऊन → का), Verbal person marker (नव्हता → नव्हतास).
D4: Varhadi	शेतात पाणी हाय का नाही जी?	शेतात पाणी आहे की नाही?	Auxiliary (हाय → आहे), Disjunctive particle (का → की), Honorific residue (जी).

it lacks aligned parallel Standard Marathi target translations required for sequence-to-sequence text normalization.

To construct a supervised normalization benchmark, native Marathi linguists performed sentence-level parallel translation of RESPIN dialectal prompts into normative Standard Marathi (मानक मराठी), assembling an initial clean parallel corpus of 16,163 pairs across agriculture and banking domains. The initial corpus comprises D1 Malvani (5,576 pairs), D2 Ahirani (5,501 pairs), and D4 Varhadi (5,086 pairs). A comprehensive manual audit protocol flagged critical translation flaws including garbled syntax (incomplete or malformed sentences), semantic inversions (meaning reversal in normalization), Hindi language leakage (Standard Hindi substituted for Standard Marathi), retained dialect residue (incompletely normalized outputs), and dropped question marks (loss of interrogative intent). All flagged pairs were manually corrected and validated before seeding the synthetic expansion pipeline.

3.3 LLM-Driven Synthetic Generation Architecture

To address data scarcity, we developed an automated synthetic parallel data generation pipeline powered by Gemma-2 [8] in both 9B-instruction-tuned (`gemma-2-9b-it`) and 27B-instruction-tuned (`gemma-2-27b-it`) configurations, accessed via the Google AI Studio API. The pipeline architecture incorporates several design decisions to ensure generation quality:

- **1-Prompt-Per-Sentence Architecture**: Rather than generating batches of parallel pairs per API call (which risks array length mismatches and cross-sentence context contamination), each dialectal sentence is processed individually with a self-contained system prompt.
- **Zero-Hallucination System Prompt**: The generation prompt enforces strict semantic fidelity rules: the standard output must preserve exact meaning, domain terminology (agriculture/banking), named entities, and numerical values from the dialectal input. The prompt includes dialect-specific lexical mapping tables and 5–8 few-shot reference examples per dialect.
- **Temperature Control**: Generation temperature is set to 0.3 with top-p = 0.9 to balance diversity with fidelity. Multiple candidate generations are produced and ranked by verification scores.
- **Domain Preservation**: System prompts explicitly enumerate 100% preservation rules for domain-specific entities including crop names (शेती, कापूस, ज्वारी), banking terminology (कर्ज, खाते, व्याजदर), and geographical references.

Below is the self-contained System Prompt template deployed for Gemma-2 synthetic parallel data generation:

SYSTEM PROMPT (Gemma-2 Dialect Normalization Teacher)

```
You are an expert Marathi Computational Linguist specializing in
regional sub-dialects of Maharashtra (Malvani, Ahirani, Varhadi).

YOUR TASK: Normalize the given dialectal Marathi input into
grammatically perfect, pure Standard Marathi (मानक मराठी).

STRICT RULES:
1. Preserve 100% of original sentence meaning. DO NOT add, remove,
or alter facts.
2. Maintain all domain-specific terminology
(Agriculture/Banking/Finance).
3. Do NOT substitute Marathi words with Hindi or English
equivalents.
4. Output ONLY valid Devanagari text string in Standard Marathi.

FEW-SHOT EXAMPLES:
[Input Dialect]: मका आज बँकेंत जावचा हा.
[Standard Output]: मला आज बँकेत जायचे आहे.

[Input Dialect]: मना पोराले शेतीमा काम करणा शे.
[Standard Output]: माझ्या मुलाला शेतीत काम करायचे आहे.

INPUT SENTENCE: {dialect_sentence}
STANDARD MARATHI TRANSLATION:
```

3.4 Multi-Tier Verification and Quality Filtering

All generated pairs undergo a rigorous three-tier verification and correction pipeline:

1. **Tier 1: Rule-Based Syntactic Filter.** Automated validation checks implement Indic Abugida script normalization principles [2], including:
   – *Devanagari Script Integrity*: Both source and target must contain >90% Devanagari Unicode characters within the `U+0900--U+097F` range, automatically normalizing broken nuktas (`U+093C`), invalid halants (`U+094D`), and orphan matras.
   – *Length Ratio Constraint*: $0.5 \leq |S_{\text{dialect}}|/|S_{\text{standard}}| \leq 2.0$ to eliminate hallucinated sentence expansions or accidental truncations.
   – *Non-Identity Constraint*: $S_{\text{dialect}} \neq S_{\text{standard}}$ to exclude trivially unnormalized identical string copies.
   – *Minimum Token Count*: Both source and target must contain ≥ 3 whitespace-separated tokens.
2. **Tier 2: LLM Semantic Verification.** A secondary LLM evaluator using `llama-3.1-8b-instant` (accessed via Groq API with key rotation pool and NVIDIA NIM fallback for rate limit resilience) performs semantic equivalence verification. The verifier rates candidates on a 1–5 score scale assessing:

Table 2. Parallel Dataset Composition and Yield Across All Three Dialect Suites

Suite ID	Dialect Name	Original Pairs	Synthetic Pairs	Total Pairs
D1	Malvani (Konkan)	5,576	5,569	**11,145**
D2	Ahirani (Khandesh)	5,501	5,534	**11,035**
D4	Varhadi (Vidarbha)	5,086	5,069	**10,155**
Combined	**All 3 Dialects**	**16,163**	**16,172**	**32,335**

(a) exact semantic preservation; (b) absence of hallucinated entities; (c) absence of Hindi/English contamination; and (d) complete normalization of dialectal inflections. Pairs scoring < 4.0 are rejected or routed to correction.

3. **Closed-Loop Correction Engine.** Pairs flagged by Tier 1 or Tier 2 enter a two-step correction loop: Step 1 generates corrected candidate translations using feedback-aware prompts highlighting the specific failure reason (e.g., "Hindi leakage detected" or "Incomplete normalization"); Step 2 (QA Auditor) re-verifies corrected candidates, discarding pairs that fail a second audit pass.

As detailed in Table 2, the synthetic augmentation pipeline contributed 16,172 high-quality verified parallel pairs, nearly perfectly doubling the original corpus to produce an expanded dataset of 32,335 samples. The near-1:1 ratio between original and synthetic pairs across all three dialects demonstrates the consistency and reliability of the generation pipeline.

3.5 Evaluation Partitioning Strategy

To ensure rigorous, unbiased evaluation and prevent data leakage during hyperparameter selection, all datasets are partitioned using a strict 85/15 stratified split protocol:

– **15% Held-Out Test Set**: Completely isolated *prior* to any model training. This partition is reserved exclusively for final metric reporting and is never exposed to the model during training or validation. Stratification ensures proportional representation of dialect labels and domain categories.
– **85% Training Pool**: The remaining data is used for model training with 5-Fold Cross-Validation (80% train / 20% validation per fold). Fold-level validation metrics are used for hyperparameter selection and early stopping decisions.

This protocol is applied identically across all 8 dataset configurations: D1-16k, D2-16k, D4-16k, D124-16k (combined), and their 32k expanded counterparts.

3.6 Cascaded Speech-to-Standard-Text Pipeline Architecture

To achieve robust real-world speech processing for non-standard Marathi dialects, our framework deploys a two-stage cascaded architecture linking speech recognition with text normalization, as illustrated in the system flow below:

CASCADED SPEECH-TO-TEXT NORMALIZATION PIPELINE

[**Dialectal Audio Input**] *(Malvani D1 / Ahirani D2 / Varhadi D4)*

$\Downarrow$

[**Stage 1: ASR System**] $\longrightarrow$ `ai4bharat/indic-conformer-600m-multilingual` *(RNN-T, Baseline WER: 23.71%)*

$\Downarrow$

[**Stage 2: Raw Transcript**] *(Phonetic dialectal text with non-standard suffixes)*

$\Downarrow$

[**Stage 3: Normalizer**] $\longrightarrow$ `google/mT5-small` *(300M, 250k Vocab, Fine-Tuned, Output WER: 16.58%)*

$\Downarrow$

[**Standard Output**] $\longrightarrow$ Normative Standard Written Marathi (मानक मराठी) *(Rel. WER Drop: −30.07% vs ASR)*

Fig. 1. Cascaded Architecture: Linking Acoustic ASR Transcriptions with Sequence-to-Sequence Text Normalization.

3.7 Subword Tokenization Breakdown: IndicBART (64k) vs. mT5-Small (250k)

A fundamental factor driving model performance discrepancies is subword tokenization granularity. Table 3 illustrates subword segmentation differences between IndicBART's 64,000-token SentencePiece tokenizer and mT5's 250,000-token tokenizer on a representative Ahirani dialect input (मनाले बँकमा क्रेडिट कार्ड काढण्यासाठी शेतस).

As shown in Table 3, IndicBART fragments the dialectal pronoun मनाले into three isolated subword pieces (_मन, _ा, _ले), destroying the stem-suffix boundary and forcing the decoder to reconstruct grammar from fragmented character chunks. In contrast, mT5's 250k vocabulary represents मनाले and बँकमा as single intact tokens, enabling direct sequence-level mapping to Standard Marathi मला and बँकेत.

3.8 IndicConformer Baseline ASR Evaluation

To establish a comprehensive speech-and-language baseline, we evaluate pre-trained automatic speech recognition (ASR) performance on the official `IISc_RESPIN_test_mr` test benchmark set, comprising 2,170 speech utterances (3.04 hours of recorded audio) across all four regional sub-dialects: D1 (Southern Konkan/Malvani), D2 (Northern Konkan/Ahirani), D3 (Standard Pune Marathi), and D4 (Varhadi/Vidarbha).

Table 3. Subword Tokenization Granularity: IndicBART (64k) vs. mT5-Small (250k)

Tokenizer / Model	Subword Token Sequence
Dialect Input Text	मनाले बँकमा क्रेडिट कार्ड काढण्यासाठी शेतस
IndicBART Tokenizer (64k)	[_मन, _ा, _ले, _बँक, _मा, _क्रेडिट, _कार्ड, _काढ, _ण्या, _साठी, _शे, _त, _स]
Fragment Count	13 subword tokens (Excessive fragmentation of dialectal inflections)
mT5-Small Tokenizer (250k)	[_मनाले, _बँकमा, _क्रेडिट, _कार्ड, _काढण्यासाठी, _शेतस]
Fragment Count	**6 whole tokens** (Preserves intact morpho-syntactic boundaries)

Table 4. Baseline ASR Performance (`indic-conformer-600m`) on `IISc_RESPIN_test_mr`

Decoder Mode	Utterances	Duration (h)	Norm. WER (%)	Norm. CER (%)	Exact Match (%)
CTC Decoding	2,170	3.04	23.94%	5.12%	30.12%
RNN-T Decoding	2,170	3.04	**23.71%**	**5.06%**	**30.78%**

We evaluate `ai4bharat/indic-conformer-600m-multilingual` under both Connectionist Temporal Classification (CTC) and Recurrent Neural Network Transducer (RNN-T) decoding setups. As detailed in Table 4, the RNN-T decoder achieves optimal baseline performance with 23.71% Word Error Rate (WER), 5.06% Character Error Rate (CER), and 30.78% Exact Match Accuracy.

Sub-dialect analysis reveals significant acoustic-phonetic difficulty variation across regional varieties:

- **D3 (Standard Pune Marathi)**: Achieves baseline ASR accuracy of 10.87% WER and 2.34% CER (555 utterances).
- **D2 (Ahirani)**: Achieves 24.14% WER and 5.08% CER (540 utterances).
- **D4 (Varhadi)**: Achieves 24.79% WER and 5.14% CER (516 utterances).
- **D1 (Malvani)**: Proves to be the most challenging dialect for speech recognition, exhibiting 34.37% WER and 7.45% CER (559 utterances) due to distinct vocalic shortening and phonetic contractions.

3.9 Evaluation Metrics

All results are reported using standardized automatic evaluation metrics computed via SacreBLEU [20] and Word Error Rate (WER) scripts:

- **BLEU** (Bilingual Evaluation Understudy): Corpus-level BLEU-4 with 4-gram precision, brevity penalty, and smoothing method “exp”.
- **chrF++**: Character n-gram F-score augmented with word unigrams and bigrams, capturing subword morphological variations in agglutinative Marathi text.
- **Normalized WER / CER**: Word Error Rate and Character Error Rate computed after Devanagari Unicode script standardization.

Table 5. Comprehensive Evaluation Matrix on `IISc_RESPIN_test_mr` (2,170 Utterances)

Model Configuration	Training State	Utts	IB BLEU	IB WER (%)	mT5 BLEU	mT5 WER (%)
Non-Finetuned Base	Zero-Shot	2,170	3.12	704.22%	2.45	> 500.00%
D1 Malvani (16k)	Fine-Tuned	559	57.80	26.60%	**43.86**	**34.37%**
D2 Ahirani (16k)	Fine-Tuned	540	90.13	6.46%	**79.46**	**11.95%**
D4 Varhadi (16k)	Fine-Tuned	516	83.59	10.19%	**74.81**	**14.73%**
D124 Combined (16k)	Fine-Tuned	2,170	62.98	30.13%	**72.18**	**17.23%**
D1 Malvani (32k)	Fine-Tuned	559	24.06	60.32%	**44.36**	**32.75%**
D2 Ahirani (32k)	Fine-Tuned	540	65.41	28.70%	**79.76**	**12.61%**
D4 Varhadi (32k)	Fine-Tuned	516	67.97	25.43%	**76.90**	**14.19%**
D124 Combined (32k)	Fine-Tuned	2,170	**76.50**	**16.23%**	**73.48**	**16.58%**

4 Findings

4.1 Comprehensive Test Benchmark on `IISc_RESPIN_test_mr`

Table 5 presents the official test evaluation matrix evaluated across 2,170 utterances on the `IISc_RESPIN_test_mr` test set, comparing non-finetuned zero-shot base models, fine-tuned IndicBART (16k original vs. 32k expanded), and fine-tuned google/mT5-small (16k original vs. 32k expanded).

Key observations from Table 5 include:

1. **Failure of Zero-Shot Base Models**: Non-finetuned base models (`IndicBART` and `mT5-small`) fail completely in zero-shot mode, exhibiting severe repetitive token hallucination and producing degenerate outputs ($>$ 500% WER, BLEU $<$ 3.5). Supervised fine-tuning is strictly required for task alignment.
2. **mT5-Small Consistency**: The 32k fine-tuned `mT5-small` model achieves 73.48 BLEU, 16.58% WER, and 87.70 chrF++ on the combined multi-dialect test set. On Standard Marathi (D3), mT5-small achieves near-perfect exact alignment with **97.39 BLEU** and **1.37% WER**.
3. **IndicBART Scaling**: Fine-tuning IndicBART on the 32k combined corpus reduces its WER from 30.13% (16k) to **16.23%** (32k) with **76.50 BLEU**, demonstrating that synthetic data expansion activates latent capacity in mBART-50 architectures.

4.2 Dialectwise Relative WER Reductions vs. ASR Baseline and Raw Input

Table 6 compares the Word Error Rate of the baseline ASR system, raw dialect input text, and our fine-tuned `mT5-small` 32k model across sub-dialects.

Our fine-tuned text normalizer achieves dramatic relative WER reductions over initial ASR output:

– **Ahirani (D2)**: Reduces WER from 24.14% (ASR) / 42.35% (Raw Text) down to **12.61%**, representing a −**47.76% relative WER reduction** over ASR and −**70.22%** over raw un-normalized text.

Table 6. Dialectwise Relative WER Reductions: ASR Baseline vs. Fine-Tuned mT5-Small (32k)

Sub-Dialect	ASR Baseline WER	Raw Text WER	mT5 (32k) WER	mT5 (32k) BLEU	Rel. WER
D1 (Malvani)	34.37%	51.14%	**32.75%**	44.36	−
D2 (Ahirani)	24.14%	42.35%	**12.61%**	79.76	−4
D3 (Standard)	10.87%	0.00%	**1.37%**	**97.39**	**−8**
D4 (Varhadi)	24.79%	23.93%	**14.19%**	76.90	−4
Overall Combined	**23.71%**	**40.00%**	**16.58%**	**73.48**	**−3**

Table 7. Benchmark Performance Matrix: 16k Single-Suite Datasets

Dataset	Total Pairs	Test Set	IB BLEU	mT5 BLEU	Δ BLEU	IB chrF++	mT5 chrF++	Δ chrF++
D1 (Malvani)	5,576	836	48.52	46.31	−2.21	78.49	74.43	−4.06
D2 (Ahirani)	5,501	825	47.29	**60.31**	+13.02	66.84	**77.34**	+10.50
D4 (Varhadi)	5,086	762	72.87	**80.99**	+8.12	85.72	**91.21**	+5.49
D124 Comb.	16,163	2,424	48.17	**63.29**	+15.12	68.58	**81.37**	+12.79

– **Varhadi (D4)**: Reduces WER from 24.79% (ASR) down to **14.19%** (**−42.76% relative reduction**).
– **Standard Marathi (D3)**: Reduces WER from 10.87% (ASR) down to **1.37%** (**−87.40% relative reduction**), maintaining near-zero error degradation.
– **Overall Combined**: Reduces overall multi-dialect WER from 23.71% (ASR) / 40.00% (Raw Text) down to **16.58%** (**−30.07% relative reduction** vs. ASR and **−58.55%** vs. raw text).

4.3 Benchmark Results on 16k Original Datasets (16,163 Parallel Pairs)

Table 7 presents the performance matrix on 5-fold cross-validated 16k single-suite benchmarks. mT5-small achieves 63.29 BLEU and 81.37 chrF++ on the combined D124 16k dataset (+15.12 BLEU over IndicBART).

4.4 Benchmark Results on 32k Expanded Datasets (32,335 Parallel Pairs)

Table 8 presents the performance matrix on 32k single-suite benchmarks. Single-dialect fine-tuned mT5 models achieve **65.10 BLEU** on D1 Malvani 32k (+17.85 over IndicBART) and **62.07 BLEU** on D2 Ahirani 32k (+21.56 over IndicBART).

4.5 Qualitative Prediction Analysis

Table 12 presents representative prediction outputs from fine-tuned models on held-out test sentences from `IISc_RESPIN_test_mr`.

Table 8. Benchmark Performance Matrix: 32k Single-Suite Datasets

Dataset	**Total Pairs**	**Test Set**	**IB BLEU**	**mT5 BLEU**	**Δ BLEU**	**IB chrF++**	**mT5 chrF++**	**Δ chrF++**
D1 (Malvani)	11,145	1,671	47.25	**65.10**	+17.85	64.69	**82.88**	+18.19
D2 (Ahirani)	11,035	1,655	40.51	**62.07**	+21.56	62.21	**79.29**	+17.08
D4 (Varhadi)	10,155	1,523	73.62	**78.89**	+5.27	86.75	**90.59**	+3.84
D124 Comb.	32,335	4,850	57.12	**69.51**	+12.39	75.49	**84.58**	+9.09

Table 9. Qualitative Prediction Samples from Fine-Tuned Models on Unseen Test Data

Field	**Content**
Sample 1: Interrogative Dialect Normalization (D124 Combined 32k Model)	
Dialect Input	मनाले बँकमा क्रेडिट कार्ड काढण्यासाठी शेतस, तुनाले काय कागदपत्रे लागन?
Ground Truth	मला बँकेत क्रेडिट कार्ड काढायचे आहे, तुला काय कागदपत्रे लागतील?
IndicBART Pred	मला बँकेत क्रेडिट कार्ड काढण्यासाठी आहे, तुला काय कागदपत्रे लागतील?
mT5-Small Pred	मला बँकेत क्रेडिट कार्ड काढण्यासाठी असते, तुला काय कागदपत्रे लागतील?
Evaluation	**Accurate Interrogative Normalization:** मनाले → मला, बँकमा → बँकेत, तुनाले → तुला, लागन → लागतील.
Sample 2: Financial Domain Entity Alignment (Combined 32k Model)	
Dialect Input	क्रेडिट कार्डवरून जास्तीत जास्त किती कर्ज मिळते, आणि डेबिट कार्डवरून जास्तीत जास्त किती पैसे काढू शकतो आम्ही?
Model Output	क्रेडिट कार्डवरून जास्तीत जास्त किती कर्ज मिळते, आणि डेबिट कार्डवरून जास्तीत जास्त किती पैसे काढू शकतो आम्ही?
Reference Target	क्रेडिट कार्डवरून जास्तीत जास्त किती कर्ज मिळते, आणि डेबिट कार्डवरून जास्तीत जास्त किती पैसे काढू शकतो आम्ही?
Evaluation	**Exact String Match: BLEU 100.0, chrF++ 100.0**

4.6 Impact of Synthetic Data Augmentation

Table 10 isolates the effect of dataset scaling by comparing 16k and 32k performance for each model. IndicBART benefits substantially from augmentation on the combined D124 task (+8.95 BLEU), confirming that data scarcity rather than model capacity was the primary bottleneck. mT5-small shows a more modest combined improvement (+6.22 BLEU on D124) but starts from a much higher baseline.

Critically, cross-dialect joint training in the 32k D124 configuration provides a significant boost to IndicBART's per-dialect sub-scores: D1 Malvani reaches 52.15 BLEU (+3.63 over single-dialect IndicBART baseline) and D2 Ahirani reaches 54.35 BLEU (+7.06 over single-dialect baseline), suggesting that cross-dialect transfer learning benefits models with smaller, more linguistically focused vocabularies.

An important observation is that IndicBART's single-dialect performance *decreases* for D1 and D2 when moving from 16k to 32k, while mT5 improves dramatically. This suggests that the synthetic data distribution may slightly deviate from the original distribution in ways that IndicBART's narrower vo-

Table 10. Impact of Synthetic Data Augmentation: 16k → 32k BLEU Change

Dataset	IB 16k	IB 32k	ΔIB	mT5 16k	mT5 32k	ΔmT5
D1 (Malvani)	48.52	47.25	−1.27	46.31	**65.10**	+18.79
D2 (Ahirani)	47.29	40.51	−6.78	60.31	**62.07**	+1.76
D4 (Varhadi)	72.87	73.62	+0.75	**80.99**	78.89	−2.10
D124 Comb.	48.17	57.12	+8.95	63.29	**69.51**	+6.22

Table 11. Detailed Architecture Comparison: IndicBART (244M) vs. mT5-small (300M)

Property	IndicBART (244M)	mT5-small (300M)
Base Architecture	mBART-50 (BART Enc-Dec)	T5 Encoder-Decoder
Position Encoding	Absolute sinusoidal	Relative positional bias
Vocabulary Size	64,000 SentencePiece	250,000 SentencePiece (mC4)
Pre-training Data	11 Indic + English	101 languages (mC4)
Script Approach	Devanagari-unified	Native multi-script
Target Conditioning	Required `<2mr>` token	Raw text-to-text (no prefix)
Learning Rate	5×10^{-5} (linear warmup)	5×10^{-4} (AdaFactor)
32k D124 BLEU	57.12	**69.51** (+12.39)
32k D124 chrF++	75.49	**84.58** (+9.09)

cabulary cannot accommodate, while mT5's broader subword coverage absorbs the distributional shift effectively.

4.7 Model Architecture Analysis

Table 11 provides a detailed architectural comparison between the two primary models. The 69.51 vs. 57.12 BLEU gap on the 32k combined task (+12.39 absolute) can be attributed to three key architectural differences:

1. **Vocabulary Coverage**: mT5's 250K vocabulary provides 3.9× more subword units than IndicBART's 64K, enabling finer-grained tokenization of morphologically complex Marathi dialectal forms. This is critical for dialect normalization, where subtle suffix and infix changes must be captured at the subword level.
2. **Positional Encoding**: mT5 uses relative positional biases, which generalize better to variable-length sequences typical of dialectal text where sentence length may differ substantially between dialect and standard forms.
3. **Conditioning Overhead**: IndicBART requires explicit language tokens (`<2mr>`) that add conditioning complexity, whereas mT5's pure text-to-text format enables direct mapping without task-specific architectural constraints.

Table 12. Qualitative Prediction Samples from mT5-small on Unseen Test Data

Field	Content
Sample 1: Financial Domain Entity Alignment (Combined 32k Model)	
Dialect Input	क्रेडिट कार्डवरून जास्तीत जास्त किती कर्ज मिळते, आणि डेबिट कार्डवरून जास्तीत जास्त किती पैसे काढू शकतो आम्ही?
Model Output	क्रेडिट कार्डवरून जास्तीत जास्त किती कर्ज मिळते, आणि डेबिट कार्डवरून जास्तीत जास्त किती पैसे काढू शकतो आम्ही?
Evaluation	**Exact String Match: BLEU 100.0, chrF++ 100.0**
Sample 2: Agricultural Question Normalization (Varhadi D4)	
Dialect Input	रासायनिक शेती काऊन परवडत नाही?
Model Output	रासायनिक शेती काऊन परवडत नाही?
Gold Reference	रासायनिक शेती काऊन परवडत नाही का?
Evaluation	**High Semantic Fidelity:** Varhadi interrogative काऊन correctly retained; difference is only in optional question particle का?

4.8 Qualitative Prediction Analysis

Table 12 presents representative prediction outputs from the fine-tuned mT5-small model on held-out test sentences, demonstrating the model's capability for both exact reproduction and nuanced dialectal transformation.

4.9 Taxonomy of Qualitative Error Modes

To systematically analyze remaining failure modes across the test benchmark, we categorize qualitative prediction errors into four distinct grammatical error categories, as presented in Table 13:

The financial domain example demonstrates 100% exact string match with perfect preservation of complex domain-specific entities (क्रेडिट कार्ड, डेबिट कार्ड, कर्ज, पैसे). This is notable because the input sentence contains 24 Devanagari tokens with multiple technical banking terms that the model must recognize and preserve unchanged during the normalization process.

The agricultural example illustrates nuanced behavior on Varhadi interrogative normalization. The model correctly preserves the Varhadi interrogative form काऊन ("why") and produces semantically equivalent output, differing from the gold reference only in the optional sentence-final question particle का?—a legitimate stylistic variation in Standard Marathi.

4.10 Dialect Difficulty Hierarchy

Based on the normalization metrics, we establish a clear dialect difficulty hierarchy for the three target Marathi dialects:

1. **Easiest: Varhadi (D4)**: Achieves the highest normalization scores (80.99 BLEU, 91.21 chrF++ on 16k mT5). Varhadi transformations are predominantly systematic suffix and interrogative modifications with high regularity,

Table 13. Taxonomy of Qualitative Error Categories in Neural Dialect Normalization

Error Category	Linguistic Mechanism	Representative Example
1. Dialectal Auxiliary Retention	Retaining non-standard auxiliary forms (शेतस, हाय, शे) instead of mapping to normative आहे.	*Input*: मना पोराले काम करणा शे. *Pred*: माझ्या मुलाला काम करायचे **शे**. *Gold*: माझ्या मुलाला काम करायचे **आहे**.
2. Entity Over-normalization	Erroneously modifying valid dialectal named entities or crop terms during sequence transformation.	*Input*: बाजरी कापणी कधी करायची? *Pred*: **धान्य** कापणी कधी करायची? *Gold*: बाजरी कापणी कधी करायची?
3. Disjunctive Particle Omission	Dropping optional sentence-final interrogative or disjunctive particles (का, की, जी).	*Input*: शेतात पाणी हाय का नाही जी? *Pred*: शेतात पाणी आहे की नाही. *Gold*: शेतात पाणी आहे की नाही **जी**?
4. Agglutinative Boundary Shift	Mis-segmenting stem-postposition boundaries in complex case inflections.	*Input*: बँकेत खात्यावर पैसे जमा केले. *Pred*: बँके**च्या** खात्यावर पैसे जमा केले. *Gold*: बँकेत खात्यावर पैसे जमा केले.

making them amenable to sequence-to-sequence learning even with modest training data.

2. **Moderate: Ahirani (D2)**: Shows strong improvement with mT5 (60.31 BLEU on 16k, 62.07 on 32k) but remains challenging due to heavy consonant shifts and Gujarati/Bhili lexical influence that introduce non-Marathi subword patterns.
3. **Hardest: Malvani (D1)**: Requires substantial synthetic data expansion (jumping from 46.31 to 65.10 BLEU on mT5) due to phonetic shortening, Konkani lexical overlap, and palatalization patterns that create ambiguous surface forms.

5 Conclusion

This study presents a comprehensive, reproducible framework for multi-dialect Marathi text normalization, addressing severe data scarcity through automated synthetic data expansion and establishing rigorous neural sequence-to-sequence benchmarks.

Our LLM-driven synthetic data augmentation pipeline, powered by Gemma-2 with multi-tier verification (Devanagari script filtering and LLaMA-3.1-8B semantic auditing), successfully doubled the clean parallel corpus from 16,163 to 32,335 pairs while maintaining strict semantic fidelity—a scalable paradigm for other low-resource dialect pairs. Fine-tuning sequence-to-sequence transformers on the expanded benchmark established state-of-the-art performance: google/mT5-small (300M) achieved 69.51 BLEU and 84.58 chrF++ on the 32k combined multi-dialect dataset, outperforming IndicBART (244M) by +12.39 BLEU. Individual dialect performance reached 80.99 BLEU and 91.21 chrF++ for Varhadi (D4), demonstrating that compact pre-trained sequence-to-sequence transformers can achieve high-accuracy dialect normalization.

5.1 Limitations

This work has several limitations that should be acknowledged:

1. Synthetic data generation relies on commercial LLM APIs (Gemma-2 via Google AI Studio), introducing API cost and rate limit considerations.
2. While we benchmarked three encoder-decoder models (mT5, IndicBART, IndicTrans2), decoder-only LLMs (e.g., LLaMA-3, Gemma-2) have not been evaluated for zero-shot or fine-tuned text normalization.
3. Evaluation is currently restricted to text normalization metrics (BLEU, chrF++); downstream human evaluation across diverse domain applications remains for future work.

5.2 Future Directions

Several promising directions emerge from this work:

1. **Edge Deployment via Quantization**: Model quantization (INT8/INT4) and knowledge distillation to sub-100M parameter models for on-device deployment in rural Maharashtra.
2. **Expansion to Additional Dialects**: Extending the pipeline to Deshi/Puneri, Kolhapuri, Marathwada, and other Indic language sub-dialects.
3. **Decoder-Only LLM Benchmarking**: Evaluating instruction-tuned decoder-only models for zero-shot and few-shot dialect normalization tasks.
4. **Human Evaluation**: Complementing automatic SacreBLEU metrics with expert human assessments of semantic adequacy, grammatical correctness, and naturalness.

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